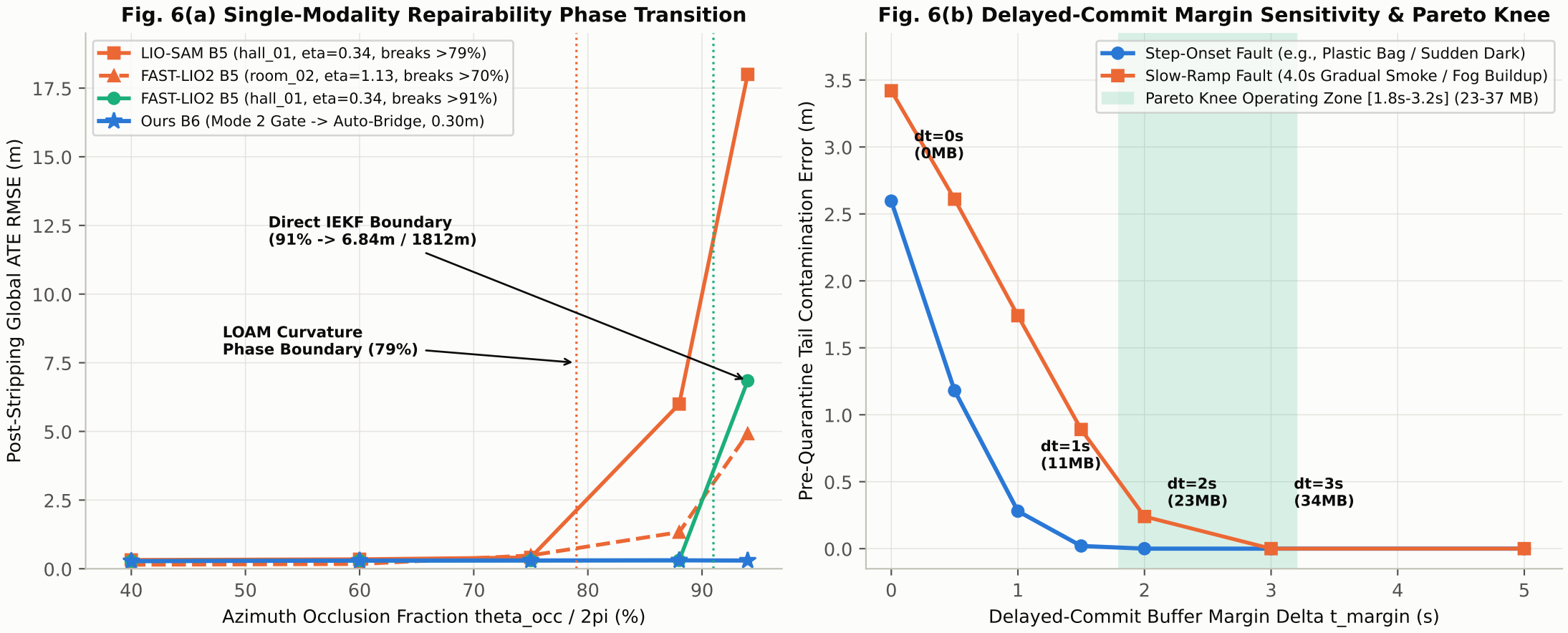
Fig. 6(a) Single-Modality Repairability Phase Transition
LIO-SAM B5 (hall_01, eta=0.34, breaks >79%)
FAST-LIO2 B5 (room_02, eta=1.13, breaks >70%)
FAST-LIO2 B5 (hall_01, eta=0.34, breaks >91%)
Ours B6 (Mode 2 Gate -> Auto-Bridge, 0.30m)
Direct IEKF Boundary
(91% -> 6.84m / 1812m)
LOAM Curvature
Phase Boundary (79%)
Post-Stripping Global ATE RMSE (m)
Azimuth Occlusion Fraction theta_occ / 2pi (%)
Fig. 6(b) Delayed-Commit Margin Sensitivity & Pareto Knee
Step-Onset Fault (e.g., Plastic Bag / Sudden Dark)
Slow-Ramp Fault (4.0s Gradual Smoke / Fog Buildup)
Pareto Knee Operating Zone [1.8s-3.2s] (23-37 MB)
dt=0s
(0MB)
dt=1s
(11MB)
dt=2s
(23MB)
dt=3s
(34MB)
Pre-Quarantine Tail Contamination Error (m)
Delayed-Commit Buffer Margin Delta t_margin (s)